

Average-weight percolation on the complete graph

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a joint work with
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Outline

Model, Results

Aldous' transition at $\frac{1}{e}$.

Ding's critical windows at criticality

Two results

Proofs

The first transition at $\frac{1}{e}$: Aldous' argument

A coupling with a branching random walk with selection

- ▶ Let K_n be the complete graph on $\llbracket 1, n \rrbracket = \{1, 2, \dots, n\}$;
- ▶ To each edge $e \in K_n$, we assign $\omega(e)$ an independent exponential r.v. with mean n ;
- ▶ A path γ is a sequence of vertices $(\gamma_0, \gamma_1, \dots, \gamma_\ell) \in \llbracket 1, n \rrbracket^{\ell+1}$ whose adjacent edges (γ_i, γ_{i+1}) are distinct;
- ▶ We call $|\gamma| = \ell$ the length of the path and $A(\gamma) := \frac{1}{\ell} \sum_{i=0}^{\ell-1} \omega(\gamma_i, \gamma_{i+1})$ the average weight of γ ;
- ▶ For $\lambda > 0$, let

$$\mathcal{L}(n, \lambda) := \max \left\{ \ell : \min_{|\gamma|=\ell} A(\gamma) \leq \lambda \right\}.$$

- ▶ Question : behaviour of $\mathcal{L}(n, \lambda)$ as $n \rightarrow \infty$?

Aldous (1998)'s transition at $\frac{1}{e}$: $\mathcal{L}(n, \lambda) := \max \left\{ \ell : \min_{|\gamma|=\ell} A(\gamma) \leq \lambda \right\}$.

Theorem (Aldous 1998)

- ▶ If $\lambda < \frac{1}{e}$, then w.h.p. $\mathcal{L}(n, \lambda) = O(\log n)$;
- ▶ If $\lambda > \frac{1}{e}$, then w.h.p. $\mathcal{L}(n, \lambda) = \Theta(n)$.

Question

What happens at criticality $\lambda = \frac{1}{e}$?

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Theorem (Ding 2013)

There are some positive constants a, b, c such that

- ▶ (critical window) if $\lambda \in [e^{-1} - (\log n)^{-2}, e^{-1} + a(\log n)^{-2}]$, then w.h.p.,

$$\mathcal{L}(n, \lambda) = \Theta((\log n)^3);$$

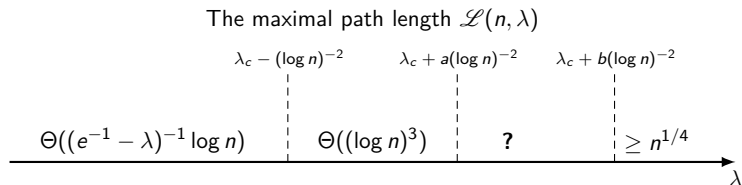
- ▶ (supercritical window) if $\lambda \geq e^{-1} + b(\log n)^{-2}$, then w.h.p.,

$$n^{1/4} \leq \mathcal{L}(n, \lambda) \leq c(\lambda - e^{-1})n;$$

- ▶ (subcritical window) if $\lambda \leq e^{-1} - (\log n)^{-2}$, then w.h.p.,

$$\mathcal{L}(n, \lambda) = \Theta\left((e^{-1} - \lambda)^{-1} \log n\right).$$

Ding's critical window sizes at criticality $\lambda_c = e^{-1}$



Questions :

- ▶ In the critical window, does the ratio $\frac{\mathcal{L}(n, \lambda)}{(\log n)^3}$ converge?
- ▶ In the supercritical window, is $n^{1/4}$ the correct rate?
- ▶ What is the precise location of the transition between the critical and supercritical windows?

Theorem (Ding and Goswami (2015))

Let $\lambda = e^{-1} + \varepsilon$ with $\varepsilon > 0$. There are some constants $c, c', \varepsilon_0 > 0$ such that for all $\varepsilon < \varepsilon_0$, w.h.p.,

$$e^{-c\varepsilon^{-1/2}} \leq \frac{\mathcal{L}(n, \lambda)}{n} \leq e^{-c'\varepsilon^{-1/2}}.$$

Question :

- Can we identify the values of c and c' (as $\varepsilon \rightarrow 0$) ?

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Our results

Theorem 1 (critical regime)

For any $\alpha < \frac{\pi^2}{2}$ and $\lambda = e^{-1}(1 + \alpha(\log n)^{-2})$,

$$\frac{1}{(\log n)^3} \mathcal{L}(n, \lambda) \xrightarrow{(p)} \frac{1}{\frac{\pi^2}{2} - \alpha}.$$

Theorem 2 (nearly supercritical regime)

Let $\alpha > \frac{\pi^2}{2}$. Assume $\varepsilon \geq \beta_n > \alpha(\log n)^{-2}$ for all large n . Let $\lambda := e^{-1}(1 + \beta_n)$. As $n \rightarrow \infty$ then $\varepsilon \rightarrow 0$, w.h.p.,

$$\frac{\mathcal{L}(n, \lambda)}{n} = e^{-\frac{\pi + o(1)}{\sqrt{2\beta_n}}}.$$

1. Let for $\alpha \in \mathbb{R}$, $\lambda = e^{-1}(1 + \alpha(\log n)^{-2})$. Then w.h.p.

$$\mathcal{L}(n, \lambda) = \begin{cases} (\frac{\pi^2}{2} - \alpha + o(1))^{-1}(\log n)^3, & \text{if } \alpha < \frac{\pi^2}{2}; \\ n^{1 - \frac{\pi}{\sqrt{2\alpha}} + o(1)}, & \text{if } \alpha > \frac{\pi^2}{2}. \end{cases}$$

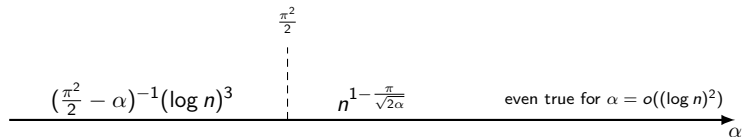
2. Let $\lambda = e^{-1}(1 + \varepsilon)$. Then w.h.p. ($n \rightarrow \infty$ then $\varepsilon \rightarrow 0$),

$$\mathcal{L}(n, \lambda) = n e^{-(\frac{\pi}{\sqrt{2}} + o(1))\varepsilon^{-1/2}}.$$

3. “ $\implies$ ” asymptotic behaviours of $\min_{|\gamma|=\ell} A(\gamma)$ depending on whether $\ell \sim c(\log n)^3$ or $\ell = ne^{-\pi/\sqrt{2\beta_n}}$.

The critical window sizes at criticality, with $\lambda = e^{-1}(1 + \alpha(\log n)^{-2})$

The second-order transition of $\mathcal{L}(n, \lambda)$

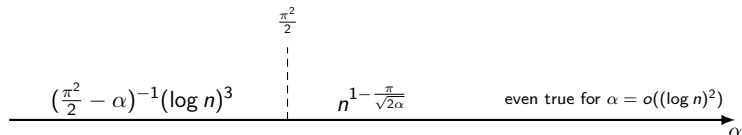


Plan :

1. e^{-1} : why does the (first-order) transition occur at e^{-1} ? [Aldous' argument]
2. Coupling : a branching random walk with selection and an exploration process in the complete graph.

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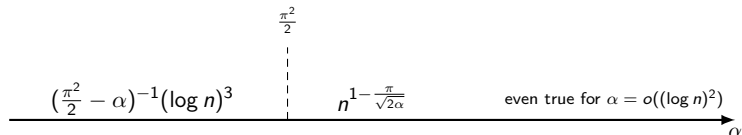


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Aldous (1998) : The upper bound. Let $\lambda < \frac{1}{e}$. To show that $\mathcal{L}(n, \lambda) = O(\log n)$ w.h.p.

- ▶ Let $\mathcal{P}_\ell$ be the collection of paths of length ℓ . Then $\#\mathcal{P}_\ell \leq n^{\ell+1}$.
- ▶ By union bound,

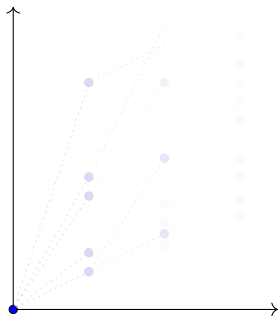
$$\begin{aligned} & \mathbb{P}(\mathcal{L}(n, \lambda) \geq C \log n) \\ & \leq \sum_{\ell \geq C \log n} \mathbb{P}(\exists \gamma \in \mathcal{P}_\ell : S(\gamma) \leq \lambda \ell) \\ & \leq \sum_{\ell \geq C \log n} n^{\ell+1} \mathbb{P}(\text{Poisson}(\frac{\lambda \ell}{n}) \geq \ell) \\ & \lesssim \sum_{\ell \geq C \log n} n (\lambda e)^\ell \\ & \rightarrow 0 \end{aligned}$$

for sufficiently large C .

Aldous (1998) : The lower bound. Let $\lambda > \frac{1}{e}$. To show that $\mathcal{L}(n, \lambda) = \Theta(n)$ w.h.p.

- ▶ Observation. For any given $i \in K_n$, $\sum_{j \neq i} \delta_{X(i,j)}$ converges locally in law to PPP(1) ($\sum_{i=1}^{\infty} \delta_{\{\tau_i\}}$, a Poisson point process of intensity 1 on $\mathbb{R}_+$).
- ▶ Idea : Exploring the graph around a given vertex locally approximates a Poisson branching random walk V .

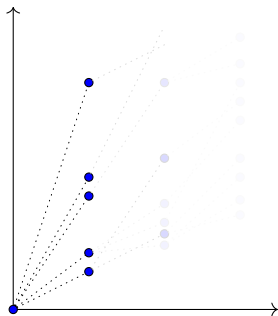
Branching random walk $(V(u))_{u \in \mathbb{T}}$



$\sum_{|u|=1} \delta_{\{V(u)\}} \stackrel{(d)}{=} \text{PPP}(1) :$
Poisson point process on $\mathbb{R}_+$.

At each generation, vertices have independently children with positions at distance a copy of $\text{PPP}(1)$ from their parent.

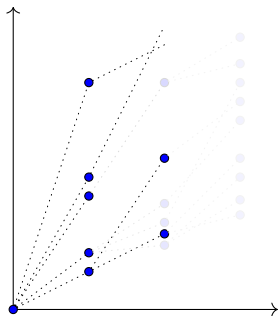
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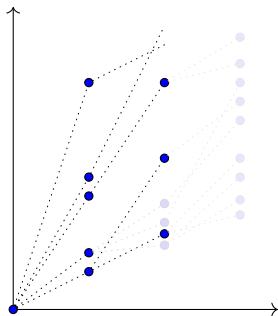
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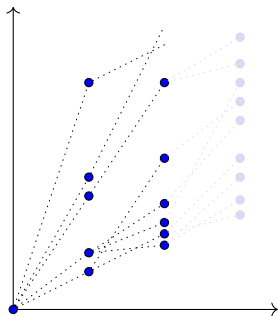
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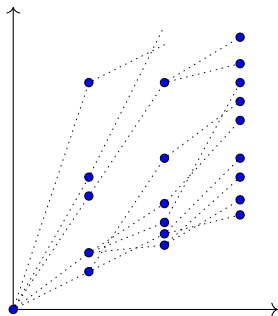
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The lower bound when $\lambda > \frac{1}{e}$

Hammersley 74, Kingman 75 and Biggins 76

Let $\varphi(t) := \log \mathbb{E} \sum_{|u|=1} e^{-tV(u)}$ for $t \in \mathbb{R}$. Almost surely

$$\lim_{n \rightarrow \infty} \frac{1}{n} \min_{|u|=n} V(u) = - \inf_{t > 0} \frac{\varphi(t)}{t} =: \nu.$$

Consequence

- ▶ For Poisson(1) BRW, $\nu = \frac{1}{e}$;
- ▶ For $\lambda > \nu$, up to $k = c(\lambda)n$, explore the graph by comparison with a Poisson(1) BRW;
- ▶ Hence w.h.p., $\mathcal{L}(n, \lambda) \geq c(\lambda)n$.

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- ▶ Hence w.h.p., $\mathcal{L}(n, \lambda) \geq c(\lambda)n$.

- ▶ We suppose now that the weights $X(i, j)$, $i, j \in K_n$, are i.i.d. copies of $ne\text{Exp}(1) - 1$. Then $\mathcal{L}(n, \lambda) = \mathcal{L}_{\text{org}}(n, (\lambda + 1)/e)$.
- ▶ Theorem 1' : If $\alpha < \frac{\pi^2}{2}$, then as $n \rightarrow \infty$,

$$\frac{1}{(\log n)^3} \mathcal{L}\left(n, \frac{\alpha}{(\log n)^2}\right) \xrightarrow{(p)} \frac{1}{\frac{\pi^2}{2} - \alpha}.$$

Proof of Theorem 1', why " $\frac{\pi^2}{2}$ "?

Lemma 1 (a variant of Mogulskii's estimate)

Let $(Y_n)_{n \geq 1}$ be a centered one-dimensional random walk with unit variance. Let $c > 0$ be an arbitrary constant. For all large Δ and $L \gg \Delta^2$, we have

$$\mathbb{P}\left(Y_L \leq c\Delta, \max_{0 \leq i \leq j \leq L} (Y_i - Y_j) \leq \Delta\right) = e^{-(\frac{\pi^2}{2} + o(1))\frac{L}{\Delta^2}}.$$

Remark

$$\mathbb{P}\left(\max_{0 \leq i \leq j \leq L} (Y_i - Y_j) \leq \Delta\right) = e^{-(\frac{\pi^2}{8} + o(1))\frac{L}{\Delta^2}}.$$

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Proof of Theorem 1' : The upper bound in Theorems 1' ($\alpha < \frac{\pi^2}{2}$)

- ▶ Claim (Ding) : if there is a path of length greater than ℓ with average weight below λ , then there is a path of length in $[\ell, 2\ell]$ with average weight below λ .
- ▶ Let $\lambda = \frac{\alpha}{\ln^2 n}$, $t > \frac{1}{\frac{\pi^2}{2} - \alpha}$, and $\ell = \lfloor t \ln^3 n \rfloor$.
- ▶ We have

$$\begin{aligned} & \mathbb{P}(\mathcal{L}(n, \lambda) \geq \ell) \\ & \leq \mathbb{P}\left(\cup_{L=\ell}^{2\ell} \{\exists \gamma \in \mathcal{P}_L : S_L(\gamma) \leq \lambda L\}\right) \\ & \leq \sum_{L=\ell}^{2\ell} \mathbb{P}\left(\exists \gamma \in \mathcal{P}_L : S_L(\gamma) \leq \lambda L, \max_{0 \leq i \leq j \leq L} (S_i(\gamma) - S_j(\gamma)) < b_n\right) + o(1) \\ & \leq \sum_{L=\ell}^{2\ell} n e^{\lambda L} \mathbb{P}\left(Y_L \leq \lambda L, \max_{0 \leq i \leq j \leq L} (Y_i - Y_j) < b_n\right) + o(1), \end{aligned}$$

where $b_n := \log n + \log \log n$, Y denotes a random walk with step distribution $\text{Exp}(1) - 1$, and the last inequality is due to a change of measures.

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Proof of the upper bound in Theorem 1'. Continuation ($\alpha < \frac{\pi^2}{2}$)

- By Lemma 1 (Mogulskii's type estimate),

$$\mathbb{P}\left(\mathcal{L}(n, \lambda) \geq \ell\right) \leq \sum_{L=\ell}^{2\ell} n e^{\lambda L} e^{-\left(\frac{\pi^2}{2} + o(1)\right) \frac{L}{(\log n)^2}} + o(1).$$

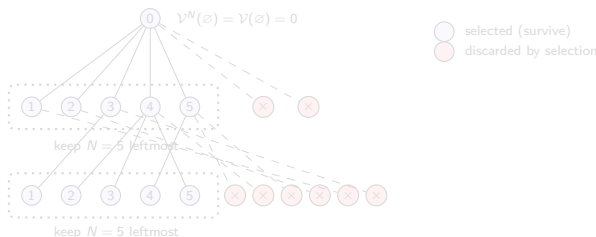
- Recall : $\lambda = \frac{\alpha}{\log^2 n}$, $t > \frac{1}{\frac{\pi^2}{2} - \alpha}$ and $\ell = \lfloor t \log^3 n \rfloor$.

- We get

$$\mathbb{P}\left(\mathcal{L}(n, \lambda) \geq \ell\right) \rightarrow 0.$$

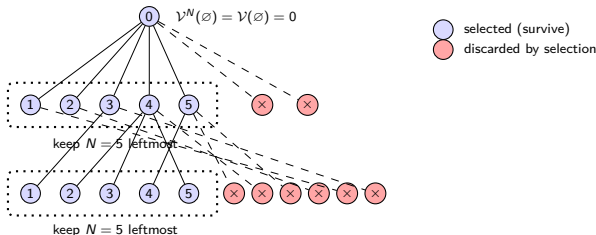
Lower bound : branching random walk with selection

- ▶ Consider a branching random walk $\mathcal{V}$ on $\mathbb{R}$. In our setting, the offspring, together with their displacements, are i.i.d. copies of $\sum_{i=1}^{\infty} \delta_{\{eT_i-1\}}$ where $0 < T_1 < T_2 < \dots$ are the atoms of a Poisson point process of intensity 1 on $\mathbb{R}_+$.
- ▶ Let N be an integer. We denote by $\mathcal{V}^N$ the branching random walk $\mathcal{V}$ in which, at each step, we only keep the N leftmost particles of the next generation.



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Brunet–Derrida’s conjecture, proved by Bérard–Gouéré (2010)

Let $v = (a.s.) \lim_{n \rightarrow \infty} \frac{1}{n} \min_{|u|=n} \mathcal{V}(u)$ and
 $v_N = \lim_{n \rightarrow \infty} (a.s.) \frac{1}{n} \min_{|u|=n} \mathcal{V}^N(u)$. Then

$$v_N - v \sim \frac{c}{(\log N)^2}$$

[In our setting, $v = 0$ and $c = \frac{\pi^2}{2}$]

N -branching Brownian motion...

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Proposition

Fix $a \in [0, 1]$ and $t > 0$. If $\mathcal{V}^N$ starts with $\lfloor N^a \rfloor$ particles located at 0, then for any sequence ℓ_N satisfying $\frac{\ell_N}{(\log N)^3} \rightarrow t$,

$$\frac{1}{\log N} \min_{|u|=\ell_N} \mathcal{V}^N(u) \xrightarrow{(L^2)} -a + \frac{\pi^2}{2}t, \quad N \rightarrow \infty.$$

- ▶ Consider $(X_i)_{1 \leq i \leq n-1}$ a family of $(n-1)$ i.i.d. copies of $X = n\text{Exp}(1) - 1$. Its increasing order statistics $X_{(0)} := -1 < X_{(1)} < \dots < X_{(n-1)}$ can be coupled with (T_i) : for all $i \in \llbracket 1, n-1 \rrbracket$,

$$eT_i - 1 \leq X_{(i)} \leq \frac{n}{n-i}(eT_i - 1) + \frac{i}{n-i}.$$

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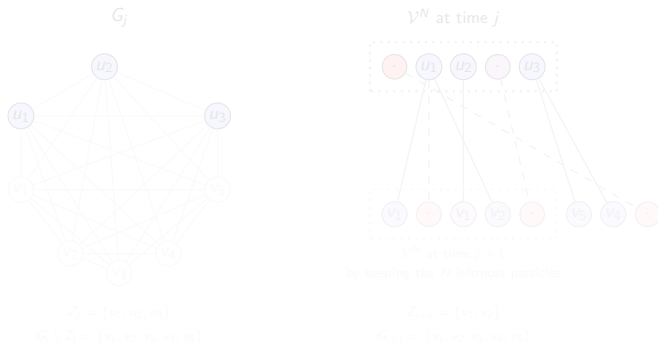
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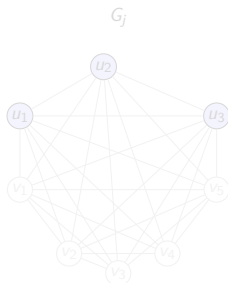
- ▶ An active vertex $u \in Z_j$ corresponds to a **blue** particle u of $\mathcal{V}^N$ at time j .
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- ▶ **Purple and red** particles at time j give birth only to **red** particles.



- ▶ If several blue particles correspond to the same $v \in G_j \setminus Z_j$, keep the leftmost one **blue** and recolor the others **purple**.
- ▶ Finally, we remove Z_j and all incident edges to obtain the graph G_{j+1} .

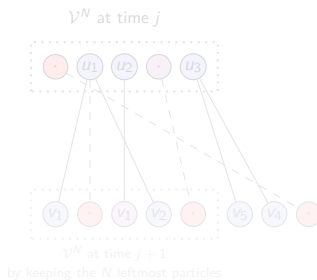
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- ▶ **Purple** and **red** particles at time j give birth only to **red** particles.



$$Z_j = \{u_1, u_2, u_3\}$$

$$G_j \setminus Z_j = \{v_1, v_2, v_3, v_4, v_5\}$$



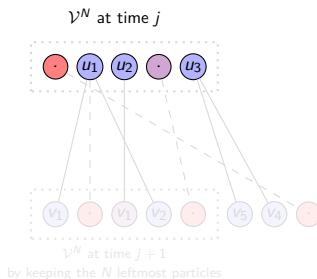
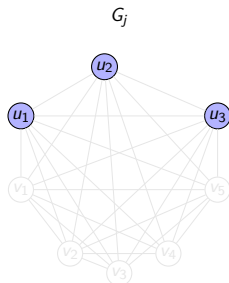
$$Z_{j+1} = \{v_1, v_2\}$$

$$G_{j+1} = \{v_1, v_2, v_3, v_4, v_5\}$$

- ▶ If several blue particles correspond to the same $v \in G_j \setminus Z_j$, keep the leftmost one **blue** and recolor the others **purple**.
- ▶ Finally, we remove Z_j and all incident edges to obtain the graph G_{j+1} .

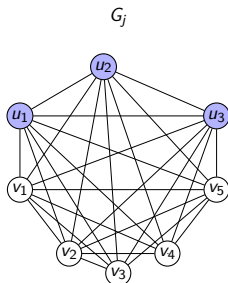
A coupling between $\mathcal{V}^N$ and an exploration process (Z_j, G_j) on K_n .

- ▶ An active vertex $u \in Z_j$ corresponds to a blue particle u of $\mathcal{V}^N$ at time j .
- ▶ Among the $n - 1$ leftmost children of u , exactly $\#G_j - \#Z_j$ are chosen uniformly and colored blue (edges to $G_j \setminus Z_j$); the others are red.
- ▶ Purple and red particles at time j give birth only to red particles.



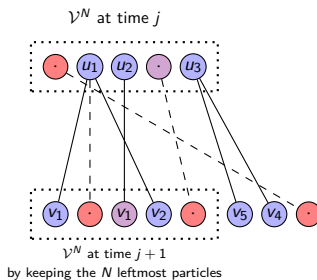
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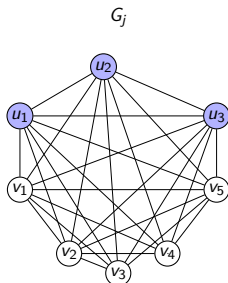
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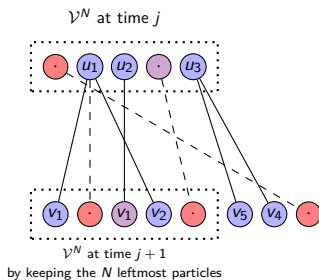
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Lemma 2

Let $\alpha < \frac{\pi^2}{2}$ and $t < \frac{1}{\frac{\pi^2}{2} - \alpha}$. Choose $a \approx 1$. Apply the above exploration process with $G_0 = K_n$ and $N_0 = N = \lfloor n^a \rfloor$. There exists, w.h.p., an active vertex at time $\lfloor t(\log n)^3 \rfloor$ such that $V(u) \leq \alpha t(\log n)$.

Corollary

With probability tending to 1,

$$\mathcal{L}(n, \frac{\alpha}{(\log n)^2}) \geq \lfloor t(\log n)^3 \rfloor.$$

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Thank you for your attention !